StockSync

Real-Time Automated Retail Inventory Tracking

on NVIDIA Jetson Orin Nano Using YOLOv8 Instance Segmentation

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Problem & Motivation

$1T

Lost annually to
retail out-of-stocks

- Manual shelf counting is slow, inconsistent, and error-prone
- Out-of-stocks lead directly to lost sales and poor customer experience
- Computer vision can automate tracking in real-time on affordable edge hardware

StockSync addresses this with a complete edge AI pipeline.

Dataset

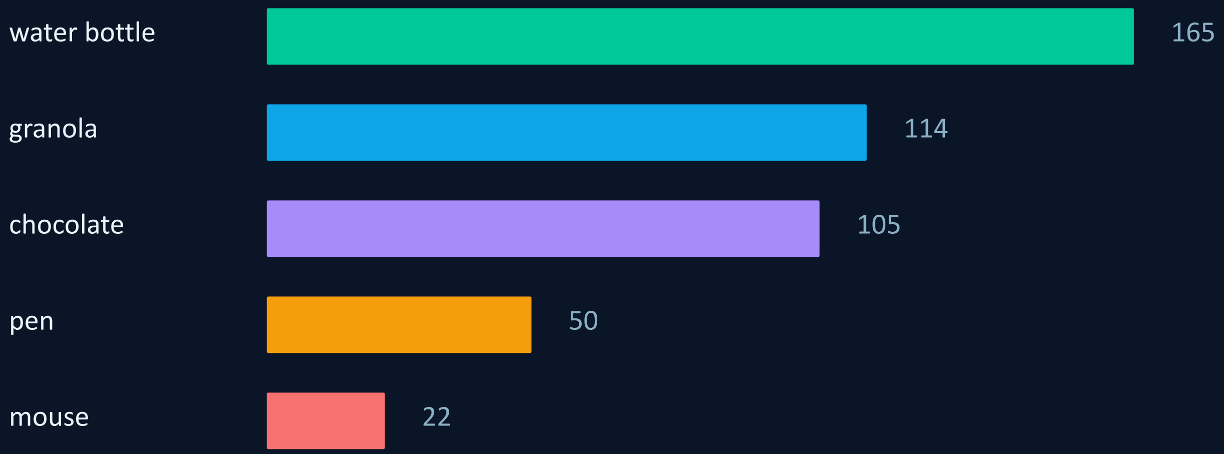
425
Images

5
Classes

Roboflow
Platform

Masks
Annotation Type

Class Distribution (total images)



Train / Val / Test Split

Class	Train	Validation	Test
water bottle	113	29	23
granola	87	15	12
chocolate	68	21	16
pen	33	8	9
mouse	18	2	2

Model & Training

YOLOv8n-seg

Parameters	**3.26M**
GFLOPs	**11.5**
Backbone	**CSP + PANet neck**
Base Weights	**ImageNet Pretrained**
Task	**Instance Segmentation**

Training Setup

Platform:	**Google Colab (Tesla T4 GPU)**
Epochs:	**65 (early stopped, patience=15)**
Best Epoch:	**Epoch 50 | Batch: 8 | Image: 640×640**
Optimizer:	**AdamW (lr=0.001111, wd=0.0005)**
Augmentation:	**Flip, HSV, Mosaic, Erase, Blur, CLAHE**

Results

88.7%

Mask mAP@50
(Test Set)

Per-Class Performance (Mask mAP@50)

Key Insight: Performance correlates with training data size. Water bottle, granola and chocolate had consistent test results, while pen and mouse ranged from 60-99% on test data. Pen (60.1%) and water bottle(98.5%) show the range. More data for pen would close the gap.

Performance correlates with training data size — more data → higher accuracy.

TensorRT Optimization

3.93× Speedup

TensorRT FP16 vs PyTorch GPU · measured over 100 frames

PyTorch (GPU)

63.35 ms

Latency

15.8 FPS

Baseline on Jetson Orin Nano

TensorRT FP16

16.11 ms

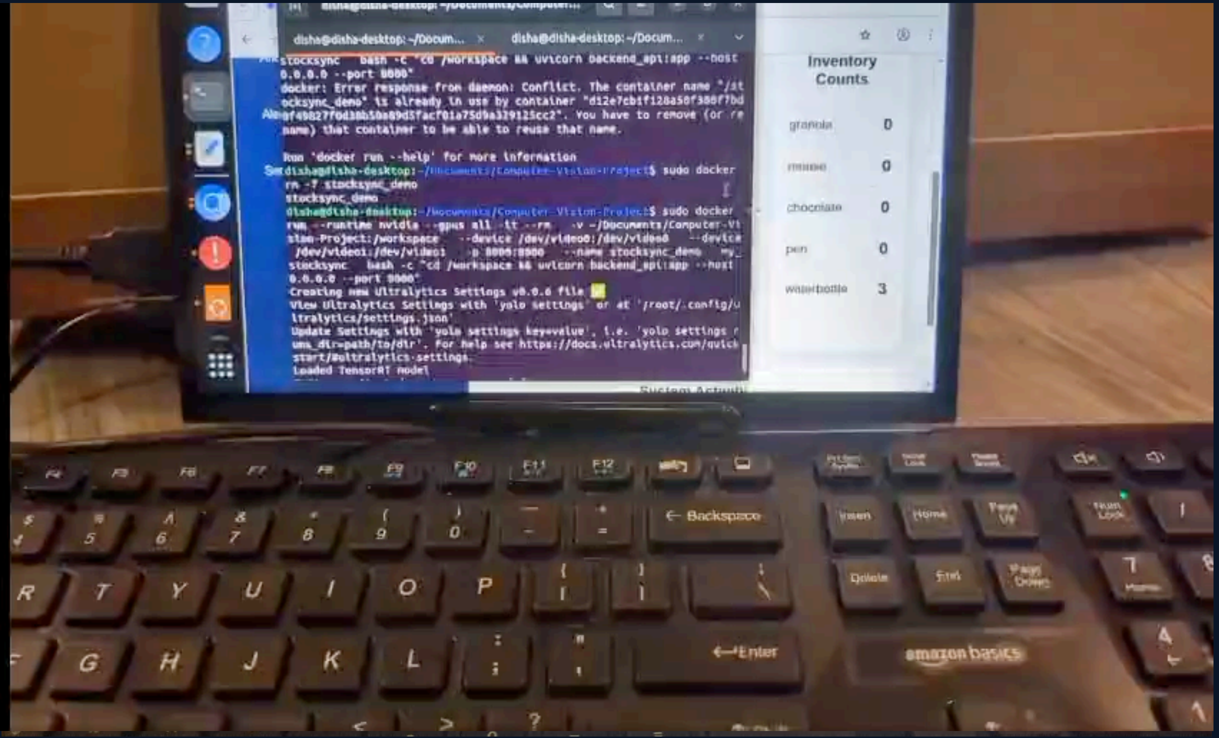
Latency

62.1 FPS

Optimized on Jetson Orin Nano

Exported best.pt → TensorRT FP16 on Jetson Orin Nano — nearly 4× faster inference at real-time frame rates.

System Demo



🎥 Live Feed

Real-time YOLOv8 segmentation overlay

📊 Inventory Counts

Per-class item count updated in real-time

🔔 Low Stock Alerts

Automatic alerts when items run low

⚡ Stack

FastAPI backend + React frontend

Conclusion & Future Work

What We Built

- ✓ Complete end-to-end computer vision pipeline
- ✓ YOLOv8 instance segmentation trained on custom retail dataset
- ✓ TensorRT FP16 optimization: 3.93× speedup on Jetson Orin Nano
- ✓ Live FastAPI + React dashboard with inventory tracking & alerts
- ✓ 88.7% Mask mAP@50 on held-out test set

StockSync — Smarter Shelves. Real-Time Intelligence.

References

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